

AP Precalculus — Unit 1 Practice

Polynomial & Rational Functions (Domain, Rates of Change, Inverses, Transformations, Asymptotes)

Name: _____

Date: _____

Unit 1 Multiple Choice

Choose the best answer for each question.

1. Which option below shows the domain of the function $f(x)$?

$$f(x) = \frac{1}{\sqrt{x} - \sqrt{2}}$$

A. $[0, \infty)$

C. $[0, 2) \cup (2, \infty)$

B. $(0, \infty)$

D. $[2, \infty)$

2. Find the average rate of change of the function $g(x)$ from $x = 2$ to $x = 4$.

$$g(x) = \frac{x}{x^2 - 1}$$

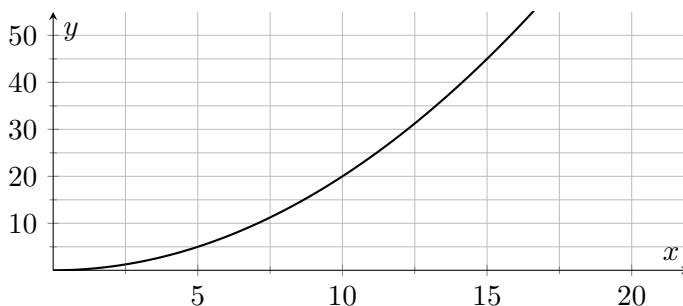
A. $-\frac{1}{2}$

C. $-\frac{1}{3}$

B. $-\frac{3}{5}$

D. $-\frac{1}{5}$

3. A function $f(x)$, when plotted on a graph, looks as shown. What is $f^{-1}(20)$ equal to?



A. 15

C. 7.5

B. 10

D. 12.5

4. If $h(x) = x^2 - 5$, then what is the value of $h(h(x))$?

A. $x^4 - 20x^2 + 20$

C. $x^4 - 10x^2 + 20$

B. $x^4 - 10x^2 + 25$

D. $2x^4 - 10x^2 + 20$

5. The function $h(x) = x^2$ is moved horizontally by 3 units to the right, then stretched vertically by a factor of 2, and then moved vertically up by 6 units. Which of the following equations shows the function after the transformations?

A. $h'(x) = 2(x - 3)^2 + 12$

C. $h'(x) = 2(x + 6)^2 - 3$

B. $h'(x) = (2x - 3)^2 + 6$

D. $h'(x) = 2(x - 3)^2 + 6$

6. $f(x) = 2x + 7$ and $g(x) = ax - b$. What is the value of $a + b$ if $f(x)$ and $g(x)$ are inverses of each other?

A. 3

C. 2

B. 4

D. 1

7. Work out the vertical asymptotes for the function:

$$\frac{x - 3}{x^2 - 5x - 14}$$

A. $x = -2$ and $x = -7$

C. $x = -2$ and $x = 3$

B. $x = -2$ and $x = 7$

D. $x = 3$ and $x = 5$

8. Work out the intervals on which

$$\frac{1}{x} < \frac{2}{2x - 9}.$$

A. $(-\infty, 1) \cup (4, \infty)$

C. $(-\infty, 0) \cup (\frac{9}{2}, \infty)$

B. $(0, \frac{9}{2})$

D. $(-\infty, 0) \cup (5, \infty)$

Unit 1 Free Response Questions

U1-FR1. Let

$$f(x) = \frac{\sqrt{x - 1}}{x - 4}.$$

- State the domain of f in interval notation.
- Explain (in one sentence) why $x = 4$ is excluded.

U1-FR2. Let

$$g(x) = \frac{x^2 + 1}{x - 1}.$$

Compute the average rate of change of g from $x = 2$ to $x = 5$.

U1-FR3. Consider

$$r(x) = \frac{x^2 - 9}{x^2 - 5x + 6}.$$

- Factor and simplify $r(x)$ completely.
- State any **holes** (removable discontinuities), including the x -value.
- State the **vertical asymptote(s)**.

U1-FR4. Let $f(x) = 3x - 8$.

- Find $f^{-1}(x)$.
- Verify your inverse by showing $f(f^{-1}(x)) = x$.

AP Precalculus — Unit 2 Practice

Exponential & Logarithmic Functions (Properties, Solving, Inverses, Composition)

Unit 2 Multiple Choice

Choose the best answer for each question.

1. Which of the exponential functions below are identical?

$$A : 3^{6x-2} \qquad B : \frac{3^{6x}}{9} \qquad C : \frac{729^x}{3}$$

A. A and C

C. A , B , and C

B. B and C

D. A and B

2. If $\log_{10} A = 5$ and $\log_{10} B = 3$, then what is the value of

$$\log_{10} \sqrt{AB^3} ?$$

- A. 7
- B. 6
- C. 2
- D. 10

3. If $\log_4(x^2 + 4) = 1.5$, then what is the value of x ?

- A. $-1, 1$
- B. $-3, 3$
- C. $-2, 2$
- D. $-4, 4$

4. Two functions, $h(x)$ and $f(x)$, are defined as $\log_e x$ and $x(x + e)$, respectively. What is the value of $h(f(e^2))$?

- A. $3 + \log_e(e + 1)$
- B. $2 - \log_e(e - 1)$
- C. $3 - \log_e(e + 1)$
- D. $3 + \log_e(1 - e)$

5. Determine the inverse of the function $f(x) = \log_6(x + 8)$.

- A. $8 \cdot 6^x$
- B. $\frac{1}{8} 6^x$
- C. $6^x + 8$
- D. $6^x - 8$

Unit 2 Free Response Questions

U2-FR1. Solve for x :

$$3^{2x-1} = 27^{x-2}.$$

Give your answer as an exact number.

U2-FR2. Write as a single logarithm:

$$\log_5(x) - 2\log_5(x - 1) + \frac{1}{2}\log_5(25).$$

State any domain restrictions.

U2-FR3. A population is modeled by $P(t) = 1200(1.08)^t$, where t is in years.

- a. Find $P(5)$.

- b. Solve for t when $P(t) = 2400$ (leave your answer as a logarithm or give a decimal rounded to the nearest hundredth).

U2-FR4. Inverse + composition. Let $f(x) = 2^x - 3$.

- Find $f^{-1}(x)$.
- Compute $f^{-1}(5)$.
- Verify $f(f^{-1}(x)) = x$.

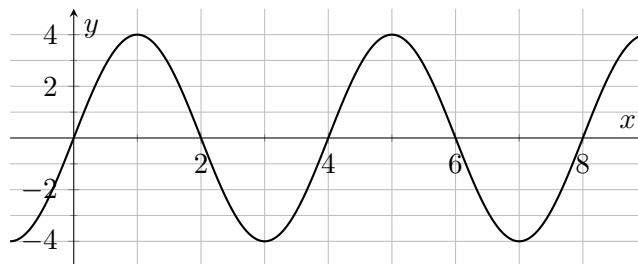
AP Precalculus — Unit 3 Practice

Trigonometric & Polar Functions (Graphing, Identities, Transformations)

Unit 3 Multiple Choice

Choose the best answer for each question.

1. The graph below shows one sinusoidal function. What are the **amplitude** and **period**?



- | | |
|------------------------------|-----------------------------------|
| A. Amplitude = 4, Period = 4 | C. Amplitude = 4, Period = 2π |
| B. Amplitude = 2, Period = 4 | D. Amplitude = 8, Period = 4 |
2. Which option shows the range of the function $y = 2(1 - 2\cos x)$?

A. $[-2, 6]$

C. $[0, 8]$

B. $[-6, 2]$

D. $[-4, 4]$

3. Two functions are defined by $f(\theta) = \sqrt{2} \sin \theta$ and $g(\theta) = \sqrt{6} \cos \theta$. If $0 < \theta < 2\pi$, for which value(s) of θ is $f(\theta) = g(\theta)$?

A. $\theta = \frac{\pi}{3}, \frac{4\pi}{3}$

C. $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$

B. $\theta = \frac{\pi}{6}, \frac{7\pi}{6}$

D. $\theta = \frac{2\pi}{3}, \frac{5\pi}{3}$

4. For some angle x , $\sin x = -\frac{\sqrt{3}}{2}$ and $\sec x = 2$. Which option shows a possible value of x (in radians) on $[0, 2\pi)$?

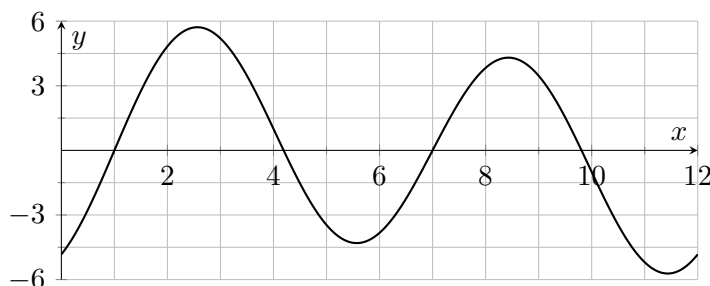
A. $\frac{11\pi}{6}$

C. $\frac{5\pi}{3}$

B. $\frac{4\pi}{3}$

D. $\frac{2\pi}{3}$

5. The graph of $g(x)$ is shown. If the function $f(x) = \frac{x}{2}$ is also drawn, for how many values of x does $f(x) = g(x)$?



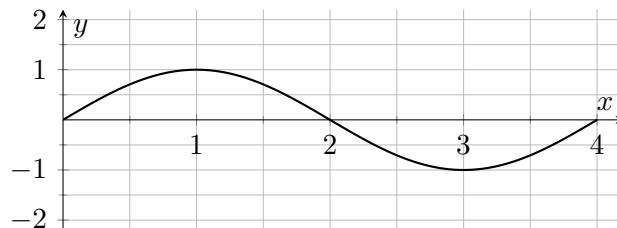
A. 1

C. 4

B. 2

D. 6

6. The graph shown represents exactly **one full cycle** of a periodic function h on the interval $0 \leq x \leq 4$. Which best describes the behavior of h on the interval $39 < x < 41$?



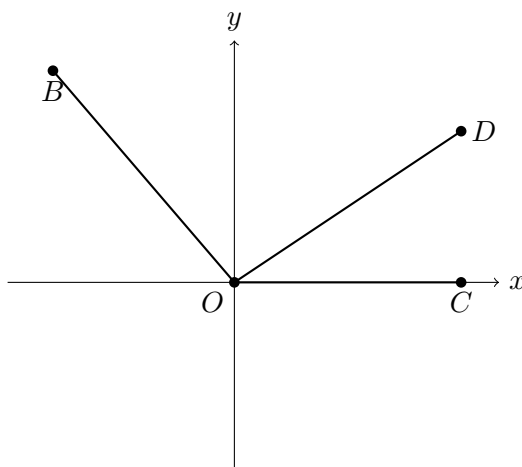
A. The function is decreasing.

C. The function is decreasing, then increasing.

B. The function is increasing.

D. The function is increasing, then decreasing.

7. The figure shows rays from the origin through points B , C , and D . Which statement is true?



- A. $\angle COB$ is in standard position with initial ray $\overrightarrow{OB}$ and terminal ray $\overrightarrow{OC}$.
 B. $\angle COB$ is in standard position with initial ray $\overrightarrow{OC}$ and terminal ray $\overrightarrow{OB}$.
 C. $\angle DOB$ is in standard position with initial ray $\overrightarrow{OB}$ and terminal ray $\overrightarrow{OD}$.
 D. $\angle DOB$ is in standard position with initial ray $\overrightarrow{OD}$ and terminal ray $\overrightarrow{OB}$.
8. Let θ be an angle in Quadrant II with $\cos \theta = -\frac{3}{5}$. What is $\sin \theta$?
- A. $\frac{4}{5}$
 B. $-\frac{4}{5}$
 C. $\frac{3}{5}$
 D. $-\frac{3}{5}$
9. The function $y = 3\sin(2x) - 5$ is a sinusoid. What is its midline?
- A. $y = 3$
 B. $y = 2$
 C. $y = -5$
 D. $y = -2$
10. What is the period of $y = \cos\left(\frac{\pi}{3}x\right)$?
- A. 3
 B. 6
 C. 2π
 D. $\frac{\pi}{3}$
11. What is the phase shift of $y = \sin\left(x - \frac{\pi}{4}\right)$?
- A. left $\frac{\pi}{4}$
 B. right $\frac{\pi}{4}$
 C. left π
 D. right π
12. Suppose $y = \sin x$ is transformed to $y = -2\sin(x) + 1$. Which statement is correct?

- A. Reflected across the y -axis; vertical stretch by 2; shifted down 1
- B. Reflected across the x -axis; vertical stretch by 2; shifted up 1
- C. Reflected across the x -axis; vertical shrink by 2; shifted up 1
- D. Reflected across the y -axis; vertical stretch by 2; shifted up 1

13. Solve on $[0, 2\pi)$: $\sin x = \frac{1}{2}$.

- A. $\frac{\pi}{6}, \frac{5\pi}{6}$
- B. $\frac{\pi}{3}, \frac{2\pi}{3}$
- C. $\frac{7\pi}{6}, \frac{11\pi}{6}$
- D. $\frac{5\pi}{3}, \frac{7\pi}{3}$

14. A sinusoidal function has a maximum value of 7 and a minimum value of -1 . What are the amplitude and midline?

- A. Amplitude = 3, midline $y = 3$
- B. Amplitude = 4, midline $y = 3$
- C. Amplitude = 3, midline $y = 0$
- D. Amplitude = 4, midline $y = 0$

15. The temperature of a city is modeled by

$$T(t) = 300 - 100 \sin\left(\frac{\pi}{45}t\right),$$

where t is a winter day ($1 \leq t \leq 90$). Which best describes the trend of $T(t)$ at $t = 22$?

- A. The temperature is falling at an increasing rate.
- B. The temperature is falling at a decreasing rate.
- C. The temperature is rising at a decreasing rate.
- D. The temperature is rising at an increasing rate.

Unit 3 Free Response Questions

U3-FR1. A sinusoidal function has midline $y = -2$, amplitude 5, and period 6.

- Write a possible equation of the form $y = a \sin(bx) + k$.
- State the maximum and minimum values of the function.

U3-FR2. Solve for all solutions on $[0, 2\pi)$:

$$2 \sin \theta = \sqrt{3} \cos \theta.$$

Give exact answers in radians.

U3-FR3. A point in the plane is given by $(x, y) = (2, -2\sqrt{3})$.

- a. Find r .
- b. Find θ in $[0, 2\pi)$.
- c. Write the polar coordinates (r, θ) .

U3-FR4. Consider the function $p(x) = \cos(x - \pi) + 2$.

- a. Identify the phase shift and vertical shift.
- b. Determine the range of $p(x)$.
- c. Find all x in $[0, 2\pi)$ such that $p(x) = 2$.